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Female Leadership Raises Aspirations and Educational Attainment for Girls: A Policy Experiment in India

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Abstract

Exploiting a randomized natural experiment in India, we show that female leadership influences adolescent girls' career aspirations and educational attainment. A 1993 law reserved leadership positions for women in randomly selected village councils. Using 8,453 surveys of adolescents aged 11–15 and their parents in 495 villages, we find that, compared to villages that were never reserved, the gender gap in aspirations closed by 25% in parents and 32% in adolescents in villages assigned to a female leader for two election cycles. The gender gap in adolescent educational attainment is erased and girls spent less time on household chores. We find no evidence of changes in young women's labor market opportunities, suggesting that the impact of women leaders primarily reflects a role model effect.

Introduction

Quotas are an increasingly common response to the enduring underrepresentation of women in various domains, from science and politics to the corporate boardroom. Around 100 countries have adopted gender quota in politics—mostly since the United Nation's Women Conference in 1995 (1)—and in 2004 Norway became the first country to mandate the presence of women on corporate boards with a 40 percent quota. Policymakers hope that quotas will have long-term effects on women's labor market outcomes over and above the immediate impact on leaders' gender balance, because the first women who become leaders may shape both parents' and children's beliefs about what women can achieve, through their policies and/or through a direct role model effect. In turn this may raise their aspirations and shape educational and career choices. We present experimental evidence from a field setting in India that supports this view.

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The idea that gender identity is at least partially a social construction is widely acknowledged in sociology (2) and social cognitive theory (3). These literatures identify belief in one's own ability (self-efficacy) as a key mechanism for personal agency, and show that this belief is highly correlated with educational aspirations and subsequent occupational choices (4). Interventions affecting these beliefs have been shown to influence long-term behavior, for example effort and performance in schools (5). Gender disparities in efficacy beliefs, in turn, are cited as an important factor behind the difference in male and female aspirations (6), especially in leadership (7, 8). Role incongruity is often emphasized as the source of the gap in beliefs (9, 10), raising the possibility that role models can challenge prevalent stereotypes and help reduce this gap. Studies show that girls may be less likely to aspire to become scientists because there are few female scientists (11–14). Exposure to own-gender experts can provide such role models, break stereotypes regarding gender roles (15) and improve individual women's aspirations and propensity to enter traditionally male-dominated areas (15–17).

Several open questions remain. These include whether role model effects can counteract potential backlash (18) and whether other social and economic constraints prevent aspirational improvements from enabling achievement gains. Further, much of the existing evidence comes from observational studies, where beliefs or actions of individuals exposed to different role models are compared (19–22), or from laboratory or short-term school-based experiments, where individuals are exposed to different role models either in person or on paper (11, 16, 23). In observational studies, people exposed to alternative role models may themselves have different preferences or opportunities. In the laboratory or school setting, data on beliefs and aspirations are often gathered after a brief exposure to a role model (24); this may not reflect real life, where exposure is more enduring. Similarly, the consequences of psychological processes on behavior are generally followed over a brief period ((5) is an exception).

In this study, we take advantage of a large-scale randomized natural experiment in India, which quickly increased the number of women in leadership positions at the village level. Exploiting a rule requiring the random selection of villages where only women could compete for this position, we compare the aspirations of parents for their sons and daughters aged 11–15, as well as the aspirations of adolescents for themselves across villages with female and male leaders. Next, we examine whether changes in girls' aspirations were accompanied by changes in educational outcomes and time spent on domestic chores. Finally, we investigate the potential channels for this effect.

Context

Since 1993, when India adopted gender quotas for elected positions on village councils, the gender balance of village leadership has dramatically altered: across India, the fraction of elected local leaders who are female has risen from under five percent in 1992 to over 40 percent by 2000 (25). In West Bengal, our area of study, as in most Indian states, one-third of village councils are randomly selected to be reserved for a woman chief councilor (or "Pradhan") in every election. Random selection of reserved councils enables us to identify the causal relationship between the election of female leaders and villagers' aspirations: there should be no other difference between villagers living in ever reserved versus unreserved councils. Furthermore, due to concurrent reservations for historically disadvantaged groups (scheduled caste and scheduled tribes), seats are not set aside to women by pure rotation: between 1998 when the system was implemented in West Bengal and 2007 when our data was collected, a village council could have been reserved for a female Pradhan once (in 1998 or 2003), twice (in 1998 and 2003), or never (26), giving us the opportunity to study the "dose response" in exposure to a female leader. Previous

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research has shown that women leaders invest in goods preferred by women, such as drinking water (27) and improves men's perception of women's leadership abilities (28).

Data

Our survey team collected data in 2006 and 2007 in one West Bengal district, Birbhum, a rural and poor district located about 200 km from Kolkata. Our survey covered a random sample of 495 villages in Birbhum's 165 village councils. Tables S1 and S2 compare 1991 census data (before reservation) and household survey data by reservation status of the council. Due to the randomization, councils have very similar characteristics. We interviewed 15 randomly selected households per village administering separate questionnaires to one prime-aged male and female adult in each household and to all adolescents (aged 11–15) residing in the household. The final sample of respondents with adolescent children includes 2,335 male and 2,438 female respondents, which covers 3,257 adolescents. Among adolescents our sample is slightly bigger, consisting of 1,852 girls and 1,828 boys (29).

The household roster had information on educational attainment. We construct three variables: 1 if ever attended school, 1 if can read and write, and last grade completed. We also report results using the normalized average across these three outcomes as dependent variables (as in (30, 31)). A time-use module asked about time spent in the last 24 hours on 16 activities covering agricultural and household activities to adults and to adolescents. We construct time spent on domestic chores as minutes spent getting water, cooking, cleaning, providing childcare, collecting fuel and doing laundry.

The adult questionnaire covered parents' aspirations for adolescents with four questions: desired educational attainment, desired age of marriage, preferred occupation at the age of 25, and whether the parent wished for the child to become Pradhan. We construct five indicator variables to capture aspirations: 1 if the parent would like the child to at least graduate from secondary school (grade 12); 1 if the parent wishes the child to marry at an age above 18; 1 if the preferred occupation is different than housewife or what the in-laws prefer, and 1 if the answer is doctor, engineer, scientist, teacher or a legal career; 1 if the parent desires the child to become the Pradhan. For adolescents, we administered the same aspiration questions, and coded them in an identical manner.

These variables are good measures of aspirations, since they are future-oriented and predictive of current behavior (4). However, since they cover separate spheres of aspirations, they could be differently affected by any intervention. Validation tests show that the first four have a fairly high degree of correlation (cronbach alpha=0.66 for parents, 0.55 for teenagers), while the wish to be Pradhan is less correlated with any of the others. Thus, in addition to the individual variables, we report the effect of the standardized average across the first four outcomes.

To investigate whether changed opportunities for young women underlie aspirational changes, we examine educational and occupational outcomes of young adults (age 16–30). Using the household roster, we construct educational variables identical to those for adolescents, and two indicator variables for occupation: 1 if not a housewife and 1 if has a high-education job, and the standardized average of the two variables. Finally, to gauge if female leadership affected schooling quality, we examine scores on a math and reading test administered to nine-year-old school-enrolled children (26).

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Results

Tables 1 and 2 present findings on how election of female leaders shapes parent and adolescent aspirations, and Figure 1 provides a graphic representation. In both tables, for each variable, Panel A presents the mean of (parents') aspirations for boys and girls in never-reserved villages, and Panel B, regression-adjusted difference in means by reservation status (26). The fifth column shows the effect across the standardized measure of the four previous outcomes, and the final column shows aspirations about becoming Pradhan.

Parents have higher aspirations for boys than for girls: on average, in villages that have never been reserved for a woman Pradhan, parents' aspirations for girls are 0.59 of a standard deviation lower than for boys (P<0.001, *t* test). For example, parents are 14 percentage points (45 percent) less likely to state that they would like their girl to graduate or study beyond the secondary school level (P<0.001, *t* test).

The gender gap in aspirations in villages with a female leader for two election cycles drops by 0.17 standard deviation (P<0.01, t test), compared to places which never had one. As parental aspirations for boys are unchanged, the entire decline in the gap is accounted for by improvements for girls. Moreover, the most significant changes occur for occupation-related aspirations. For example, the fraction of parents who believe that a daughter's, but not a son's, occupation should be determined by her in-laws declines from 76 percent to 65 percent. Previous research suggests that women have more subjective identification with women role models than men (17), implying the effect may be larger for mothers than fathers. Indeed, mothers' aspirations over education and occupation are altered more broadly while fathers primarily increase their desire for their daughters to become Pradhan (26). This last result is consistent with previous research showing that men's (but not women's) perception of women's ability to lead improved in reserved village councils (28).

Similarly, the aspirations of adolescents themselves were affected by the presence of a female leader, in the second cycle. (Figure 1, center panel, and Table 2). While boys are more ambitious than girls, the gender gap among adolescents residing in unreserved village councils is smaller than among the parents. In never-reserved villages the gender gap in adolescent aspirations is 0.41 of a standard deviation (P<0.001, *t* test). The gender gap in the desire to graduate or more is 10 percentage points (32%; P<0.001, *t* test). The decline in the gap in twice-reserved councils (average effect: 0.13 standard deviation 32%, P<0.02, *t* test) is very similar in magnitude for parents and adolescents, so girls' and boys' own aspirations remain more similar to one another than parents' aspirations for their daughters versus sons. Here again, the decline in the gap is entirely driven by an increase in girls' aspirations, not by a decrease in boys'. Adolescent girls are more likely to not want to be a housewife or have their occupation determined by their in-laws (8.3 percentage points; P=0.053, *t* test), to want to marry after 18, (8.8 percentage points; P=0.034, *t* test) and to want a job that requires an education (8.6 percentage points; P=0.02, *t* test). In the last case, the gender gap reverses.

Paralleling the changes in aspirations, the presence of female leaders also alters educational attainment and time use (Figure 1, Right panel and Table 3). Relative to girls, adolescent boys in never-reserved councils are six percent more likely to attend school (P<0.01, *t* test) and have a four percent higher likelihood of being able to read and write (P<0.01, *t* test). Like elsewhere in the developing world, adolescent girls spend more time on domestic chores than their male counterparts (79 minutes more per day in never-reserved villages; P<0.001, *t* test). The presence of female leaders did not affect educational and time-use outcomes for adolescent boys, but it improved the outcomes for girls. By the second cycle of female leadership, the gender gap in educational outcomes is completely erased (and even

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reversed), and girls spend less time on household activities (the gender gap in time spent on household activities declines by 18 minutes compared to never-reserved councils; P=0.05, *t* test)).

In all cases the gender gap starts shrinking only in the second electoral mandate held by a woman. Results in table S6 confirm that this is a "dose response" rather than a "length since first exposure" effect (26).

Women in leadership positions can change aspirations of girls through two main channels: first, by undertaking policies that make it easier for women to succeed, thus changing beliefs on what is possible for girls, and second by providing a role model of a successful woman. In any setting where actual female leadership is affected (as opposed to situations where female role models are briefly introduced (14, 32)), the two effects may coexist. No experiment can completely distinguish between the two. The trade-off is that only experiments in these settings can shed light on the impact of real-world changes in leadership.

For India, previous research has found that women leaders invest in public goods that are more in line with female preferences, in particular water infrastructure (27, 33). This may explain the reduction in time spent on domestic chores by adolescent girls. Furthermore, if female leaders invested differentially in schooling for girls, this may also underlie the change in educational outcomes we observe. We now provide several pieces of evidence, suggesting that, in this context, the role model effect of female leaders is an important channel (perhaps the most important) in changing parents' and girls' aspirations.

In Figure 2, Table S5 and (26), we find no evidence that female reservation influenced the careers, education or labor market outcomes of young adults (age 16–30), a group that adolescent girls and their parents are likely to look to when forming expectations about their future (34). The gender gap in the educational outcomes of young adults is not smaller in twice-reserved villages (P>0.6, *t* test). There is no impact on the probability that young women in villages reserved twice are housewives (P>0.8, *t* test), have jobs requiring high education (P>0.2, *t* test); have access to employment-generating schemes; P>0.5, *t* test). Finally, neither these young women (P>0.5, *t* test) nor other women in the household (P>0.4, *t* test) spend less time on domestic chores in villages.

Second, there is no evidence that female leaders made improvements in educational infrastructure that could have benefited girls more than boys. Young girls are as likely as young boys to be in school and there is no difference in tests scores (26). The reduction in the gender gap in education of adolescents is thus not due to a change in the environment. It is also probably not due to a 18 minute reduction in time spent on domestic chores. It is most likely a consequence of the change in aspirations. In turn, the reduction in time spent on chores may reflect parents giving girls time to study.

Third, the timing of the effect (starting in the second cycle) is suggestive that female leaders' image underlies the change in aspiration and educational outcomes for two reasons. First, the changes in infrastructure provision, especially greater investment in water-related facilities, occur in the first electoral cycle post reservation. In the second cycle, there is no discernible difference in infrastructure provision across once- and twice-reserved villages (27, 28). Second, previous work (28) has shown that in the first cycle, voters have a comparatively low opinion of their female leaders. However, with exposure, their beliefs change progressively: first their opinion about the ability of women to lead in general (as captured by responses to vignettes in which the gender of the politician is experimentally altered) and, by the second cycle, their opinion about their own leader's abilities. Voters in twice-reserved villages have as positive a view of their female leader as villagers in never-

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reserved areas (almost all of whom have male leaders). Voters are also more likely to elect women for unreserved seats in councils that have been reserved twice for a female leader. The timing of the effect on aspirations and education is consistent with the time it takes for female leaders to be seen in a positive light.

Discussion

Women in leadership positions have long been thought to have the potential to pave the way for long-term changes by influencing aspirations; our study demonstrates their impact in a nationwide policy experiment in India. We present evidence suggesting that this impact exceeds their (relatively limited) ability to change the concrete situation of women and girls in the short run through direct policy actions. It is their presence as positive role models for the younger generation that seems to underlie observed changes in aspirations and educational outcomes of adolescent girls.

These results show that laws can help create role models by opening opportunities that were previously unavailable to a group, and this increased opportunity does not diminish the aspirations of those outside the group. The ability of affirmative action to create role models has been subject of debate: while the policy allows members of disadvantaged groups to access positions of prestige and power, it may also cause a backlash if it creates the perception that the achievement was not due to merit. Our study shows that, in the Indian context, the positive effect of exposure to a female leader dominated any possible backlash, probably because it gave women a chance to demonstrate that they are capable leaders. And, perhaps most importantly, our study establishes that the role model effect reaches beyond the realm of aspirations into the concrete, with real educational and time-use impacts.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

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References and Notes

1. Krook, M. Quotas for Women in Politics: Gender and Candidate Selection Reform Worldwide. New York: Oxford University Press; 2009.
2. Eagly, A. Sex Differences in Social Behavior: A Social Role Interpretation. Hillsadale, NJ: 1987.
3. Bandura A. Social Cognitive Theory: An agentic perspective. Ann. Review of Psychol. 2001; 52:1–26.
4. Bandura A, Barbaranelli C, Caprara G, Pastorelli C. Self-efficacy beliefs as shapers of children's aspirations and career trajectories. Child Dev. 2001; 72:187–206. [PubMed: 11280478]
5. Cohen GL, Garcia J, Purdie-Vaugns V, Apfel N, Brzustoski P. Recursive processes in self-affirmation: intervening to close the minority achievement gap. Science. 2009; 324:400–403. [PubMed: 19372432]
6. Bussey K, Bandura A. Social cognitive theory of gender development and differentiation. Psychol. Rev. 1999; 106:676–713. [PubMed: 10560326]
7. Lips HM. College students' visions of power and possibility as moderated by gender. Psychol. of Women Quart. 2000; 24:39–43.

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8. Lips HM. Envisioning positions of leadership: the expectations of university students in Virginia and Puerto Rico. J of Soc. Issues. 2001; 57:799–813.
9. Eagly AH, Karau SJ. Role congruity theory of prejudice towards female leaders. Psychol. Rev. 2002; 109:573–598. [PubMed: 12088246]
10. Rudman LA, Phelan JE. The effect of priming gender roles on women's implicit gender beliefs and career aspirations. J. of Soc. Psychol. 2010; 41:192–202.
11. Gardiner M, Tiggemann M, Kearns J, Marshall K. Show me the money! An empirical analysis of mentoring outcomes from women in academia. Higher Educ. Res. and Dev. 2007; 26:425–442.
12. Nauta MM, Epperson DL, Kahn JH. A multiple-groups analysis of predictors of higher level career aspirations among women in mathematics, science, and engineering majors. J. of Counseling Psychol. 1998; 45:483–496.
13. Richman LS, van Dellen M. How women cope: being a numerical minority in a male-dominated profession. J. of Soc. Issues. 2011; 67:492–509.
14. Smith W, Erb T. Effect of women science career role models on early adolescents' attitudes toward scientists and women in science. J. Research in Science Teaching. 1986; 23:667–676.
15. Asgari S, Dasgupta N, Cote NG. When does contact with successful in-group members change self-stereotypes? A longitudinal study comparing the effect of quantity vs. quality of contact with successful individuals. Social Psychol. 2010; 41:202–201.
16. Lockwood P. Someone like me can be successful: do college students need same-gender role models? Psychol. of Women Quart. 2006; 30:36–46.
17. Stout JG, Dasgupta N, Hunsinger M, McManus M. STEMing the tide: Using ingroup experts to inoculate women's self-concept and professional goals in science, technology, engineering, and mathematics (STEM). J. of Personality and Soc. Psychol. 2011; 100:255–270.
18. Rudman LA, Phelan JE. The effect of priming gender roles on women's implicit gender beliefs and career aspirations. J. of Soc. Psycol. 2010; 41:192–202.
19. Nixon L, Robinson M. The educational attainment of young women: role model effects of female high school faculty. Demography. 1999; 36:185–194. [PubMed: 10332610]
20. Wright C, Huston A, Truglio R, Fitch M, Smith E, Piemyat S. Occupational portrayals on television: children's role schemata, career aspirations, and perceptions of reality. Child Dev. 1995; 66:1706–1718. [PubMed: 8556894]
21. Cane B, Rosen H. Following in her footsteps? Women's choice of college majors and faculty gender composition. Ind. and Labor Relations Rev. 1995; 48:486–504.
22. Wolbrecht C, Campbell D. Leading by Example: Female Members of Parliament as Political Role Models. Am. J. Political Science. 2007; 51:921–939.
23. Greene A, Sullivan H, Beyard-Tyler K. Attitudinal effects of the use of role models in information about sex-typed careers. J. Educational Psychol. 1982; 74:393–398.
24. For example in one of the studies in *Stout et al*, women's perceptions of their women's role in mathematics and their actual performance on a math test is analyzed after exposure to a brief interaction with a same-sex expert in the laboratory while another had laboratory participants read biographies of engineers which were randomly varied to represent women or men, and (23) examines the change in college students' self-perceptions relative to a control group after reading an article about hypothetical female or male role models.
25. Kaul S, Sahni S. Study on the participation of women in Panchayati Raj Institution. Stud. Home Comm. Sci. 2009; 3:29–38.
26. SOM Text is available as supporting material on *Science* Online.
27. Chattophadhyay R, Duflo E. Women as policy makers: evidence from a randomized policy experiment in India. Econometrica. 2004; 72:1409–1443.
28. Beaman L, Chattopadhyay R, Duflo E, Pande R, Topalova P. Powerful women: does exposure reduce bias? Q. J. Econ. 2009; 124:1497.
29. The survey team interviewed all adolescents in a household. Adult respondents were asked only about their own biological children. There were cases in which adolescents were interviewed whose biological parents were not interviewed, thereby generating a larger sample of adolescents.

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30. Kling J, Liebman J, Katz L. Experimental analysis of neighborhood effects. Econometrica. 2007; 15:83–119.

31. Karlan D, Zinman J. Microcredit in theory and practice: Using randomized credit scoring for impact evaluation. Science. 2011; 332:1278–1284. [PubMed: 21659596]

32. Nguyen T. Information, Role models, and perceived returns to education: experimental evidence from Madagascar. Mimeo MIT. 2008

33. Beaman L, Duflo E, Pande R, Topalova P. Political reservation and substantive representation: evidence from Indian village councils. India Policy Forum, Brookings and NCAER. 2010–11; 7:159–191.

34. Jensen R. Economic opportunities and gender differences in human capital: An experimental test for India. Mimeo UCLA. 2010

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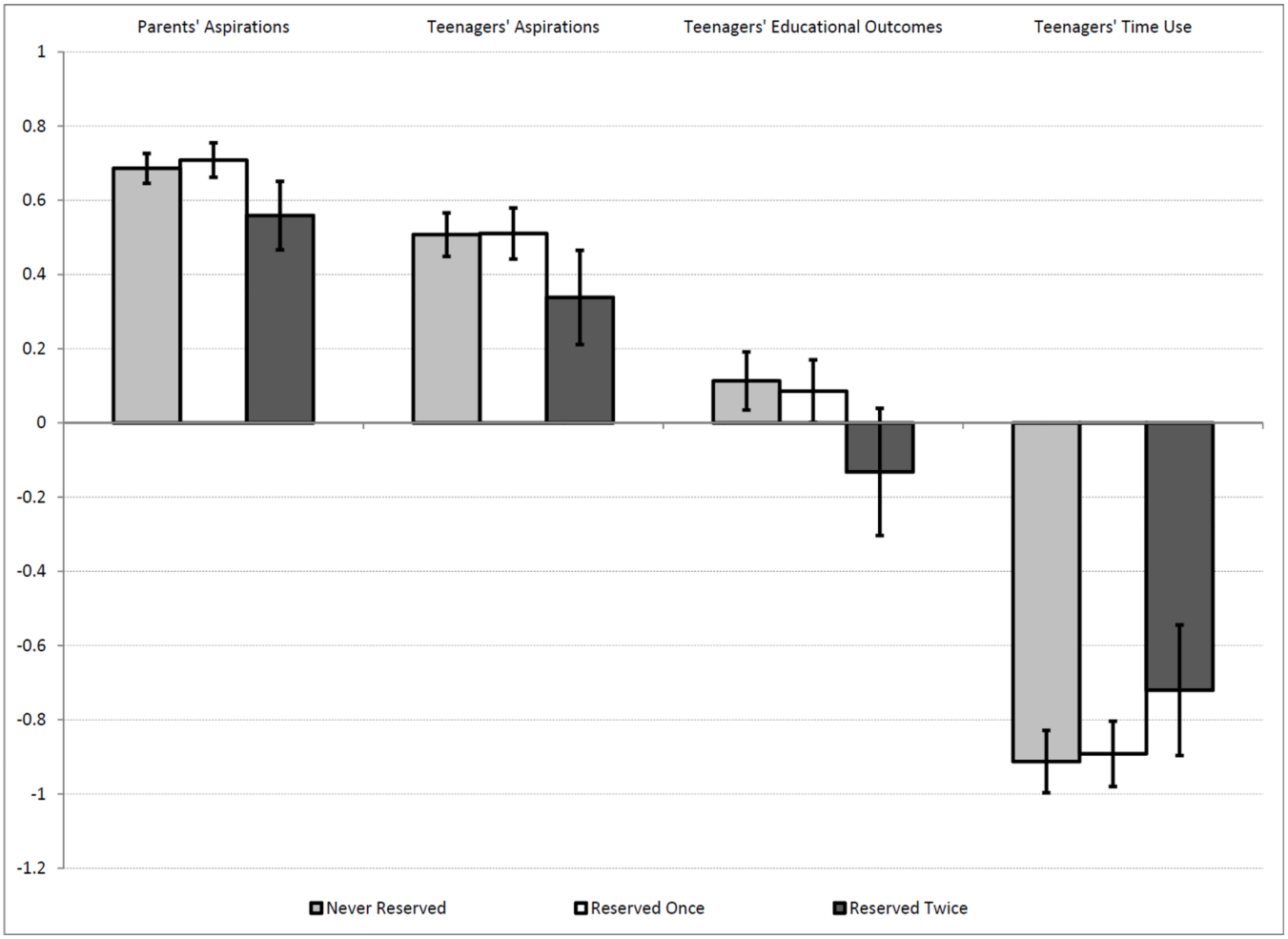

Fig. 1.

Gap in Aspirations, Educational Outcomes and Time Use.

The gray, white and black bars represent difference in the normalized average of aspirations of parents for their adolescent boys and girls (leftmost panel) and the difference in normalized average of aspirations, educational outcomes and time spent on domestic chores, between adolescent boys and girls themselves in village councils that have never been reserved, have been reserved only once and have been reserved twice for a female leader respectively. The error bars represent the 95-percent confidence interval.

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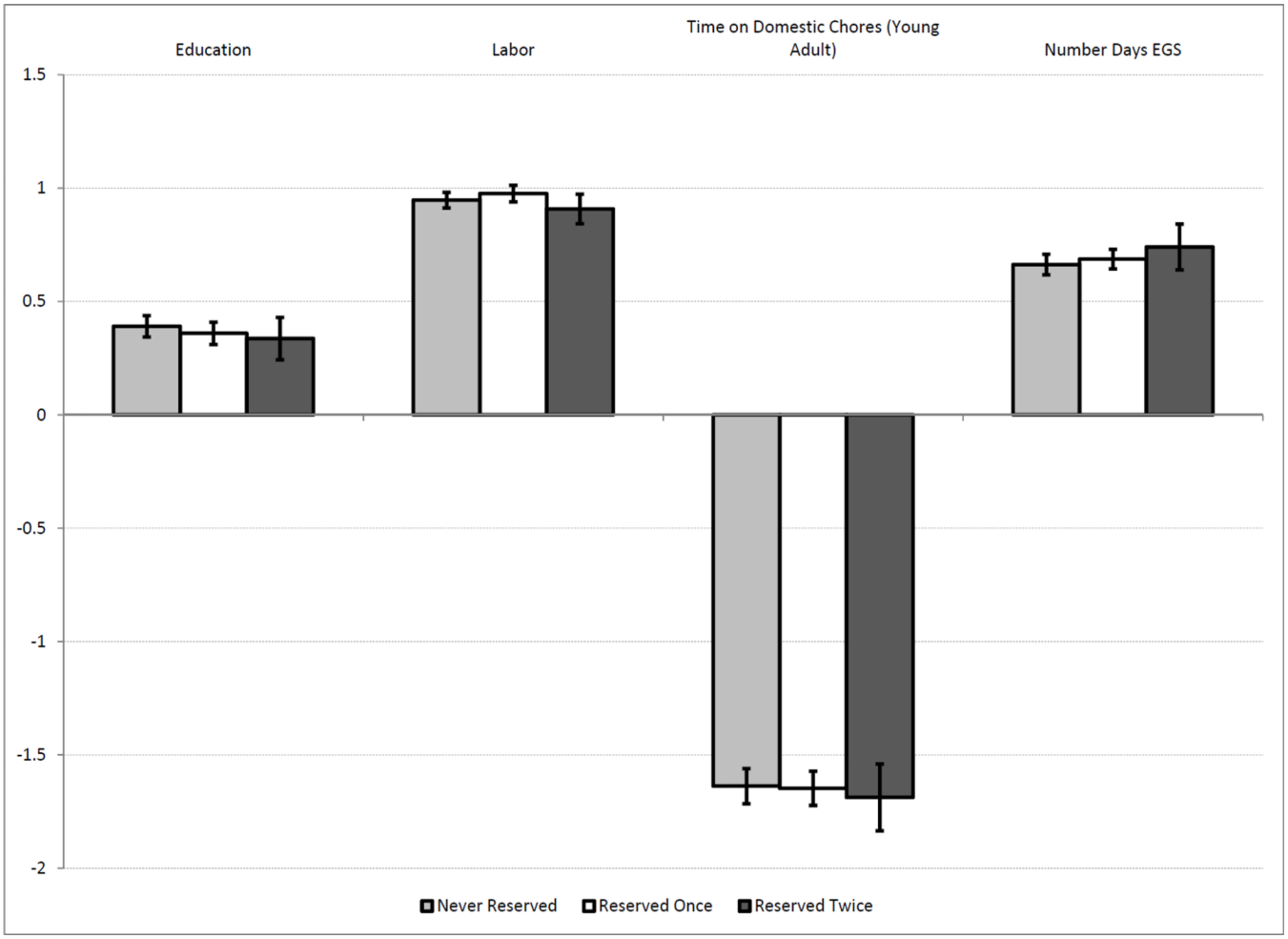

Fig. 2.

Potential Channels: Gap in Young Adult Educational Outcomes, Young Adult Labor Market Outcomes, Young Adult Time Use, and Access to Public Services

The gray, white and black bars represent the gender gap in normalized average of young adults educational and labor market outcomes, adult time use and access to employment generation schemes, in village councils that have never been reserved, have been reserved only once and have been reserved twice for a female leader respectively. The error bars represent the 95-percent confidence interval.

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Table 1

Parents' Aspirations for Their Children. Panel A shows means and standard deviations (denoted in brackets) in never reserved village councils of the variable in the column heading. The coefficients in panel B are OLS results for the dependent variable denoted in the column heading and measure the difference in aspirations for boys and girls relative to the never reserved sample. The row denoted Gap for never reserved village councils shows the gap in aspirations between boys and girls. The subsequent "Difference in Gap" rows show the gap in aspirations for boys and girls relative to the gap in the never reserved sample. For example, the last number in column 5 says that the gap in overall aspirations for girls versus boys has shrunk by 0.136 standard deviations (more than 25%) in twice reserved village councils relative to never reserved ones. Standard errors corrected for heteroskedasticity and clustering at the village council level are shown below the coefficients in parentheses. The sample size is 6140. The normalized average is computed across the four outcomes (columns 1–4), which are first normalized by subtracting the mean for never reserved village councils and dividing by standard deviation in the never reserved sample. All estimates control for village characteristics described in table S1 and for respondent gender.

	Does not wish child to be housewife or whatever in-laws prefer	Wishes child to have a high education job	Wishes child to marry after age 18	Wishes child to graduate or get higher education	Normalized average	Wishes child to be Pradhan
A. Means and SD in never reserved						
Boys	0.991	0.068	0.992	0.318	0.329	0.688
	[0.092]	[0.252]	[0.089]	[0.466]	[0.446]	[0.463]
Girls	0.242	0.039	0.756	0.176	−0.357	0.588
	[0.429]	[0.193]	[0.430]	[0.381]	[0.654]	[0.492]
B. Coefficients						
	Never Reserved					
Gap	−0.747	−0.028	−0.238	−0.140	−0.684	−0.093
	(0.018)	(0.009)	(0.012)	(0.019)	(0.025)	(0.020)
	Reserved Once					
Boys	−0.005	0.016	−0.004	0.003	0.013	−0.041
	(0.009)	(0.014)	(0.006)	(0.025)	(0.027)	(0.025)
Girls	−0.015	0.020	−0.021	−0.015	−0.011	0.000
	(0.023)	(0.009)	(0.014)	(0.020)	(0.031)	(0.029)
Difference in						
Gap	−0.010	0.003	−0.017	−0.018	−0.024	0.041
	(0.025)	(0.015)	(0.016)	(0.026)	(0.035)	(0.029)
	Reserved Twice					

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	Does not wish child to be housewife or whatever in-laws prefer	Wishes child to have a high education job	Wishes child to marry after age 18	Wishes child to graduate or get higher education	Normalized average	Wishes child to be Pradhan
Boys	−0.033	−0.012	−0.018	0.003	−0.042	0.008
	(0.018)	(0.016)	(0.010)	(0.044)	(0.039)	(0.043)
Girls	0.064	0.012	0.021	0.051	0.093	0.101
	(0.028)	(0.014)	(0.025)	(0.036)	(0.050)	(0.044)
Difference in Gap	0.097	0.025	0.038	0.048	0.136	0.092
	(0.036)	(0.020)	(0.026)	(0.048)	(0.059)	(0.049)

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Table 2

Adolescents' Aspirations. See caption to table 1 for explanation on table layout. The sample size is 3680. All estimates control for village characteristics described in table S1.

	Does not wish to be housewife or whatever in-laws prefer	Wishes to have a high education job	Wishes to marry after age 18	Wishes to graduate or get higher education	Normalized average	Wishes to be Pradhan
A. Means and SD in never reserved						
Boys	0.998	0.157	0.980	0.296	0.248	0.499
	[0.048]	[0.364]	[0.141]	[0.457]	[0.447]	[0.500]
Girls	0.600	0.166	0.660	0.195	−0.260	0.485
	[0.490]	[0.373]	[0.474]	[0.396]	[0.758]	[0.500]
B. Coefficients						
	Never Reserved					
Gap	−0.398	0.008	−0.319	−0.099	−0.507	−0.017
	(0.026)	(0.020)	(0.025)	(0.018)	(0.034)	(0.026)
	Reserved Once					
Boys	−0.008	0.053	−0.004	0.038	0.050	−0.013
	(0.011)	(0.022)	(0.013)	(0.026)	(0.029)	(0.035)
Girls	−0.031	0.062	0.010	0.035	0.045	−0.010
	(0.034)	(0.021)	(0.031)	(0.022)	(0.045)	(0.039)
Difference in						
Gap	−0.023	0.009	0.014	−0.003	−0.005	0.003
	(0.036)	(0.028)	(0.034)	(0.028)	(0.052)	(0.039)
	Reserved Twice					
Boys	0.008	−0.014	0.003	0.034	0.017	0.057
	(0.017)	(0.034)	(0.021)	(0.047)	(0.048)	(0.056)
Girls	0.083	0.086	0.088	0.035	0.182	0.049
	(0.043)	(0.036)	(0.041)	(0.035)	(0.056)	(0.055)
Difference in						
Gap	0.075	0.099	0.085	0.001	0.166	−0.008
	(0.042)	(0.051)	(0.049)	(0.045)	(0.057)	(0.053)

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Table 3

Adolescents' Educational Outcomes and Time Use. See caption to table 2 for explanation of the table layout. The sample size is 3680. All estimates control for village characteristics described in table S1.

	1 if attends school	1 if can read and write	Grade completed	Normalized average	Total time on domestic chores (minutes per day)
A. Means and SD in never reserved					
Boys	0.744	0.947	5.538	0.055	29.298
	[0.436]	[0.225]	[2.166]	[0.760]	[50.841]
Girls	0.684	0.908	5.409	−0.058	108.231
	[0.465]	[0.289]	[2.386]	[0.898]	[96.884]
B. Coefficients					
	Never Reserved				
Gap	−0.060	−0.038	−0.132	−0.112	78.954
	(0.019)	(0.014)	(0.113)	(0.041)	(4.845)
	Reserved Once				
Boys	0.005	−0.020	0.019	−0.020	−5.095
	(0.022)	(0.013)	(0.117)	(0.042)	(3.719)
Girls	0.010	−0.002	−0.002	0.004	−6.844
	(0.026)	(0.015)	(0.138)	(0.051)	(7.178)
Difference in					
Gap	0.005	0.018	−0.021	0.024	−1.748
	(0.029)	(0.021)	(0.153)	(0.058)	(6.475)
	Reserved Twice				
Boys	0.005	−0.012	−0.094	−0.025	2.869
	(0.044)	(0.031)	(0.247)	(0.100)	(6.881)
Girls	0.098	0.050	0.587	0.223	−14.772
	(0.049)	(0.029)	(0.237)	(0.102)	(10.153)
Difference in					
Gap	0.093	0.062	0.681	0.248	−17.641
	(0.051)	(0.033)	(0.243)	(0.102)	(8.971)